PP-B-002215(Army-GL)
September 27, 1984
SUPERSEDING
(See 6.4)

INTERIM FEDERAL SPECIFICATION

# BEEF STEW, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

## 1. SCOPE

1.1 Scope. This specification covers beef stew thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

| | | |
|---|---|---|
| TT-C-495 | - | Coatings, Exterior, For Tinned Food Cans |
| PPP-B-636 | - | Boxes, Shipping, Fiberboard |

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 8940

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(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Forth Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Canned Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)



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U.S. Department of Agriculture (USDA)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Standards for Grades of Frozen Peas

(Application for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Division, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Environmental Protection Agency (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Spice and Trade Association (ASTA)

Official Analytical Methods of the American Spice and Trade Association

(Application for copies should be addressed to the American Spice and Trade Association, Inc., 580 Sylvan Avenue, Englewood Cliffs, NJ 07632.)



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Association of Official Analytical Chemists (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

American Dehydrated Onion and Garlic Association (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 251 Kearney Street, No. 410, San Francisco, CA 94108.)

National Academy of Sciences

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any combination of carcasses, quarters, and any of the following recognizable primal and sub-primal cuts: square-cut chucks, shoulder clods, chuck rolls, ribs, ribeyes, trimmed full loins, trimmed short loins, strip



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loins, sirloins, top sirloin butts, rounds, top rounds, knuckles, and bottom rounds. Recognizable cuts are those which, when compared to IMPS cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. All suitable lean meat shall be used except tenderloins, and any part of a rough cut may be excluded at the option of the contractor. The beef shall be in the fresh-chilled state and shall be in excellent condition; i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class, grade and cut of meat, and typical of meat which has been properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, or microbial activity. The fat shall show no more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, the meat shall show no evidence of freezing, defrosting or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless trimmed beef shall meet the limitations specified in tables I and II.

3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the same day as boning and trimming shall be held in the temperature range of 28° to 50°F (inclusive) at all times prior to processing.

b. Holding the fresh chilled state for not more than 4 days is permitted provided that the beef is maintained in the temperature range of 28° to 40°F (inclusive) at all times prior to processing.

c. Holding raw beef in the frozen state is permitted provided that the beef is:
- placed in the freezer within 4 hours after certification and frozen to 0°F within 72 hours
- stored at 0°F or lower at all times
- held for not more than 60 days
- protected from freezer deterioration and damage



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- stored in containers that are adequate to maintain product excellence
- maintained without defrosting until time of processing into the finished product.

3.2.2 Water. Water used for ice making, formulation, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.2.3 Peas, frozen. Frozen peas shall possess uniform color, flavor, and odor and shall be free of defects such as insect damage and extraneous material. Peas shall be U.S. Grade B or better of the U.S. Standards for Grades of Frozen Peas. Alternatively, fresh peas of comparable quality may be substituted.

3.2.4 Starch, modified, high opacity. The high opacity starch shall be white, odorless, finely pulverized, modified maize food starch for use in canned foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress and conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.5 Onions, dehydrated, chopped. Chopped dehydrated onions shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.6 Salt. Salt shall be non-iodized, white, refined, sodium chloride, with or without anticaking agents.

3.2.7 Sugar. Sugar shall be clean, refined, sucrose, liquid or granulated, cane or beet sugar or both and shall comply with MIL-STD-900.

3.2.8 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.9 Hydrolyzed vegetable protein (HVP). The hydrolyzed vegetable protein shall be obtained by the hydrolysis of any one or any combination of corn, soybean or wheat proteins from which none of the monosodium glutamate has been extracted during processing and shall have a characteristic color and flavor.



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The salt-free solids shall be not less than 40 percent. The pH of a 3-percent solution shall not be less than 5.0 nor more than 6.3. Hydrolyzed protein shall be soluble in water at 90°F to the extent of not less than 5 grams in 100 milliliters.

3.2.10 Onion powder. Onion powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.11 Monosodium glutamate. Monosodium glutamate shall be odorless, free flowing, and may possess either a slightly sweet or slightly salty flavor. The crystals shall be white and translucent.

3.2.12 Paprika. Paprika shall be Spanish paprika and shall possess a bright orange to red color with a color value of not less than 110 American Spice and Trade Association (ASATA) units.

3.2.13 Celery flavoring, soluble. Soluble celery flavoring shall be prepared by properly blending and dispersing a minimum of 4 percent by weight of total extractives of the dried fruit of Apium graveolens L. on a soluble, dry, edible carrier. The flavoring shall possess a clean, celery-like aroma with a slightly bitter taste. It shall be free flowing, free from caking, free from all extraneous matter, and shall be yellowish-grey in color.

3.2.14 Black pepper, ground. The ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot biting pungent flavor and a light grey to speckled black-grey color. The volatile oil content shall be not less than 2.0 mL/100 g. and the black pepper shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.15 Potatoes. Potatoes shall be dehydro-frozen Idaho Russets with a piece size of approximately 1/2 by 1 by 1 inch. Alternatively, fresh sulfited potatoes may be used.

3.2.16 Carrots. Carrots shall be fresh or frozen canner grade and shall possess a typical bright color with a normal flavor and odor. The carrots shall be free from rot and nematode injury.



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3.2.17 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and processing.

3.3.1 Beef preparation. The boned and trimmed beef shall be cut or diced into chunks approximately 1 inch by 1 inch by 1 inch in dimension. The beef chunks shall be heated in water, until no raw portions are visible, drained and the liquid retained for use in gravy preparation. The cooked beef chunks shall be cooled to 40°F as rapidly as possible and held at this temperature for not more than 4 hours prior to filling of tray pack cans. Cooling of the beef chunks is not necessary if filling of tray pack cans occurs immediately after cooking.

3.3.2 Vegetable preparation. The carrots shall be cut into chunks approximately 1-1/2 inches in length with a diameter of not more than 1 inch. The potatoes and carrots shall be mixed in the ratio of 66.7 percent potatoes and 33.3 percent carrots.

3.3.3 Gravy preparation. The gravy shall be prepared from the following ingredients:

| Ingredients | Percent |
|---|---|
| Water and broth | 82.96 |
| Peas | 8.00 |
| Starch, modified | 5.25 |
| Onions, chopped, dehydrated | 1.27 |
| Salt | 1.00 |
| Sugar | 0.50 |
| Onion powder | 0.27 |
| Lecithin | 0.25 |
| HVP | 0.20 |
| Monosodium glutamate | 0.12 |
| Paprika | 0.10 |
| Celery flavoring, soluble | 0.04 |
| Black pepper, ground | 0.04 |



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a. A slurry shall be made with the starch and part of the water. The remaining ingredients shall be mixed together and heated to not less than 140°F. The starch slurry shall be added and the final mixture, heated to a temperature of 180° to 190°F, and held for approximately 5 minutes.

b. The volume of the final mixture shall be adjusted to compensate for evaporation loss during heating.

c. The gravy shall be used within 4 hours after cooking and held at a temperature of not less than 150°F during this interval.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with cooked beef chunks (approximately 42 ounces), potato and carrot mix (approximately 26 ounces), and gravy (approximately 38 ounces) such as to conform to the finished product requirements and to the following requirements:

a. Gravy temperature at time of filling of trays shall be not less than 150°F.
b. Beef temperature at time of filling of trays shall be 28°F to 40°F, if filling does not occur immediately after cooking.
c. Each can shall be hermetically sealed under a minimum of 10 inches (Hg) vacuum immediately after filling.
d. The filled and sealed tray pack cans shall be in the retort process within 1 hour after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at a temperature of 250°F or below until a sterilization value ($F_o$) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material, such as, but not limited to dirt, insect, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. Beef texture shall not be dry, mushy, rubbery, or tough.



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e. Vegetable texture shall not be mushy, hard, fibrous, or tough.

f. Drained weight of beef chunks shall be not less than 29 ounces.

g. Drained weight of potato pieces shall be not less than 15 ounces.

h. Drained weight of peas and carrots shall be not less than 8 ounces.

i. Total weight of fat chunks, opaque connective tissue, ligament, tendon, blood vessel, cartilage, gristle, or glandular material shall be not more than 3 ounces.

j. Beef chunks shall be discernible (not less than 75 percent distinct pieces by weight).

k. There shall be no bone greater than 0.3 inch in any dimension.

l. Gravy consistency shall be smooth, (without lumps) and shall not be excessively thick or thin.

m. The average net weight shall be not less than 106 ounces.

n. No individual can shall have a net weight of less than 104 ounces.

o. The average fat content of the finished product shall be not greater than 9.0 percent.

p. No individual tray pack shall have a fat content greater than 11.0 percent.

q. No individual tray pack shall have a salt content less than 0.5 percent or greater than 1.25 percent.

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.



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3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts. Any nonconforming beef shall be rejected.

4.4.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the Agricultural Marketing Service (AMS) agent by



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the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or presence of any off-condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

TABLE I. Sampling plan for boned and trimmed beef

| Lot size (pounds) | Sample size (No. of sample units) | Defect categories | | | |
|---|---|---|---|---|---|
| | | Major | | Minor | |
| | | AC | RE | AC | RE |
| 500 or less | 13 | 1 | 2 | 3 | 4 |
| 501 to 1200 | 20 | 2 | 3 | 5 | 6 |
| 1201 to 3200 | 32 | 3 | 4 | 7 | 8 |
| 3201 to 10000 | 50 | 5 | 6 | 10 | 11 |
| 10001 to 35000 | 80 | 7 | 8 | 14 | 15 |
| Reinspection lots of 10001 to 35000 | 125 | 10 | 11 | 21 | 22 |



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TABLE II. Boned and trimmed beef defects (cont'd)

| Category | | Defect 1/ 2/ |
|---|---|---|
| Major | Minor | |
| 151 | | Presence of lymph gland measuring 1.0 inch or more in any dimension. |
| | 201 | Presence of lymph gland measuring 0.5 inch or more but less than 1.0 inch in any dimension. |
| 152 | | Presence of blood clot measuring 1.0 inch or more in any dimension. |
| | 202 | Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension. |
| 153 | | Presence of bruise measuring 1.0 inch or more in any dimension. |
| | 203 | Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension. |
| 154 | | Presence of bone measuring 0.3 inch or more in any dimension. |
| 155 | | Presence of cartilage measuring 0.5 inch or more in any dimension. |
| 156 | | Presence of backstrap or other ligament measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other). |
| | 204 | Presence of backstrap or other ligament measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other). |



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TABLE II. Boned and trimmed beef defects (cont'd)

| Category | | Defect 1/ 2/ |
|---|---|---|
| Major | Minor | |
| 157 | | Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or gracilis membrane) measuring 2.0 square inches or more. |
| | 205 | Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or gracilis membrane) measuring 0.5 square inches or more but less than 2.0 square inches. |
| | 206 | Presence of heavy connective tissue on lower edge of short plate or flank. |
| | 207 | Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension. |
| | 208 | Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measuring 3 square inches or more. |
| 158 | | Presence of knuckle cover. |
| 159 | | Presence of kidney, pizzle eye, prepubic tendon, thymus gland or hanging tender measuring 1.0 inch or more in one dimenion and 0.2 inch or more in a second dimension (when measured at right angles to each other). |
| | 209 | Presence of kidney, pizzle eye, prepubic tendon, hanging tender or thymus gland measuring less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other). |



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TABLE II. Boned and trimmed beef defects (cont'd)

| Category | | Defect 1/ 2/ |
|---|---|---|
| Major | Minor | |
| | 210 | Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more. |
| | 211 | Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface. |
| | 212 | Presence of dehydrated surface measuring 1.0 square inch or more. |
| | 213 | Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more. |
| | 214 | Presence of exposed major blood vessel measuring 1.0 inch or more in any dimension. |
| | 215 | Presence of cod, udder, kidney, or pelvic fat. |

1/ Acceptability of product with respect to the presence of foreign material (for example: glass, dirt, insect parts, hair, wood, metal) shall be determined by a Meat and Poultry Inspection Operations employee.

2/ Evidence of freezing or defrosting or presence of any off-condition (including but not restricted to being off-odor, slightly sticky, gassy, rancid, or sour, or showing discoloration) shall cause rejection of the involved lot.



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4.4.3 Ingredient and component examination. Conformance of ingredients and components to identity, condition and other requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.6.2 Product inspection. Eight tray packs shall be selected at random from the lot. The eight tray packs shall be heated for 15 minutes in 140° to 150°F water, opened, and inspected in accordance with table III.

a. If no defects are found, the lot shall be acceptable.
b. If two or more defects are found, the lot shall be rejected.
c. If only one defect is found, a second sample of eight tray packs shall be selected at random from the lot, heated as above, opened, and inspected for the defects indicated in table III. If any defect is found in the second sample, the lot shall be rejected.



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TABLE III. Product inspection 1/ 2/

| Category | Defect |
|---|---|
| Major | |
| 151 | Drained weight of beef chunks in a can is less than 29 ounces. 3/ |
| 152 | Drained weight of potato pieces in a can is less than 15 ounces. 3/ |
| 153 | Drained weight of peas and carrots in a can is less than 8 ounces. 3/ |
| 154 | Total weight of fat chunks, opaque connective tissue, ligament, tendon, blood vessel, cartilage, gristle or glandular material weighs more than 3 ounces. |
| 155 | Beef chunks not discernible (less than 75 percent distinct pieces by weight). 4/ |
| 156 | Color not characteristic of product (materially darkened, scorched, or oxidized). |
| 157 | Beef texture dry, tough, rubbery, or mushy. |
| 158 | Vegetable texture mushy, hard, fibrous, or tough. |
| 159 | Presence of bone greater than 0.3 inch in any dimension. |
| 160 | Gravy consistency lumpy or excessively thick or thin. |

1/ The presence of any foreign material (e.g., glass, dirt, insect parts, hair, wood, metal, etc.) foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale, etc.), or foreign color shall be cause for rejection of the lot.



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2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquify rendered fat and remove gravy without breaking the beef chunks. The contents shall then be poured into a U.S. Standard 1/4 inch sieve in a manner that will distribute the product over the sieve without breaking the beef chunks. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45° angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight of the beef, potatoes, and peas and carrots shall be determined individually and shall be reported to the nearest 0.1 ounce.

4/ Percentage shall be based on drained weight.

4.6.3 Fat and salt content testing. Nine tray pack cans shall be selected from the lot and distributed as follows:

-Three for laboratory analysis.
-Three for submission to the contractor.
-Three for retention by the AMS agent as a reserve sample.

The three tray packs for laboratory analyses shall individually be tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Meat and Meat Products, Section: Meat, except that preparation of the samples shall be as follows. The three unopened cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The lot shall be rejected if:



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a. The average fat content of the three tray packs is greater than 9.0 percent.
b. The fat content of any individual tray pack is greater than 11.0 percent.
c. The salt content in any individual tray pack is less than 0.5 percent or greater than 1.25 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the fat or salt content requirement. When the reserve samples are analyzed, the analyses for both fat and salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable), in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Reinforcement with nonmetallic strapping or tape is not used; (4) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above top edge of box; (5) Length or width of pads less than box dimensions by more than 1/8 inch; (6) Arrangement or number of cans not as specified.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).



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5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product, sealed, and thermoprocessed as specified in 3.4, 3.5, and 3.6. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of grade V3c fiberboard. The pads shall be placed between the cans and on the top and bottom of the stacked cans. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s or V4s fiberboard.



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5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.7 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 35 minutes. Avoid overheating to the point where internal pressure is evident (can shows evidence of bulging). Remove from water and open lid for serving.
CAUTION: Use care when opening as pressure may have been generated within the can.

b. To heat in oven: Either punch several holes in lid of tray can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F.

WARNING: Do not place closed can in oven, as heating will increase internal pressure which may cause the can to burst.



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YIELD

Approximately 12 servings of 1 cup each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article inspection. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should include specific instructions in all procurement documents regarding arrangements for selection, inspection, and approval of the first article.



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6.4 Supersession data. This document includes the requirements of PD 401, Beef Stew, Thermostabilized, Tray Pack, dated 31 December 1980.

Preparing activity:

Army - GL

Project No. 8940-A466

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| PP-B-002215(Army-GL) | Beef Stew, Thermostabilized, Tray Pack |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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NATICK OP, 1 Apr 84

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